

universität freiburg



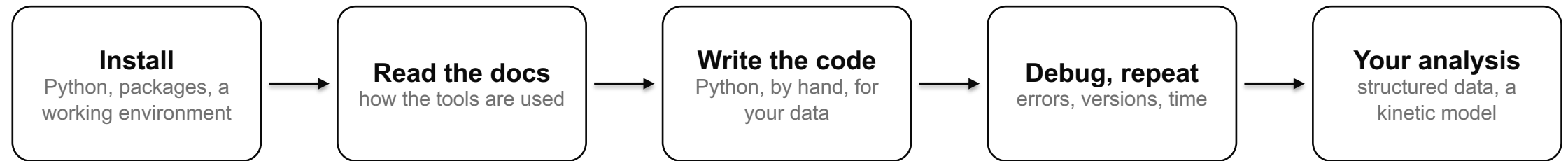
7TH ENZYMEML WORKSHOP · RÜDESHEIM · HANDS-ON SESSION 2 · 16 SEPTEMBER 2026

From measurement to kinetic modelling

Max Häußler

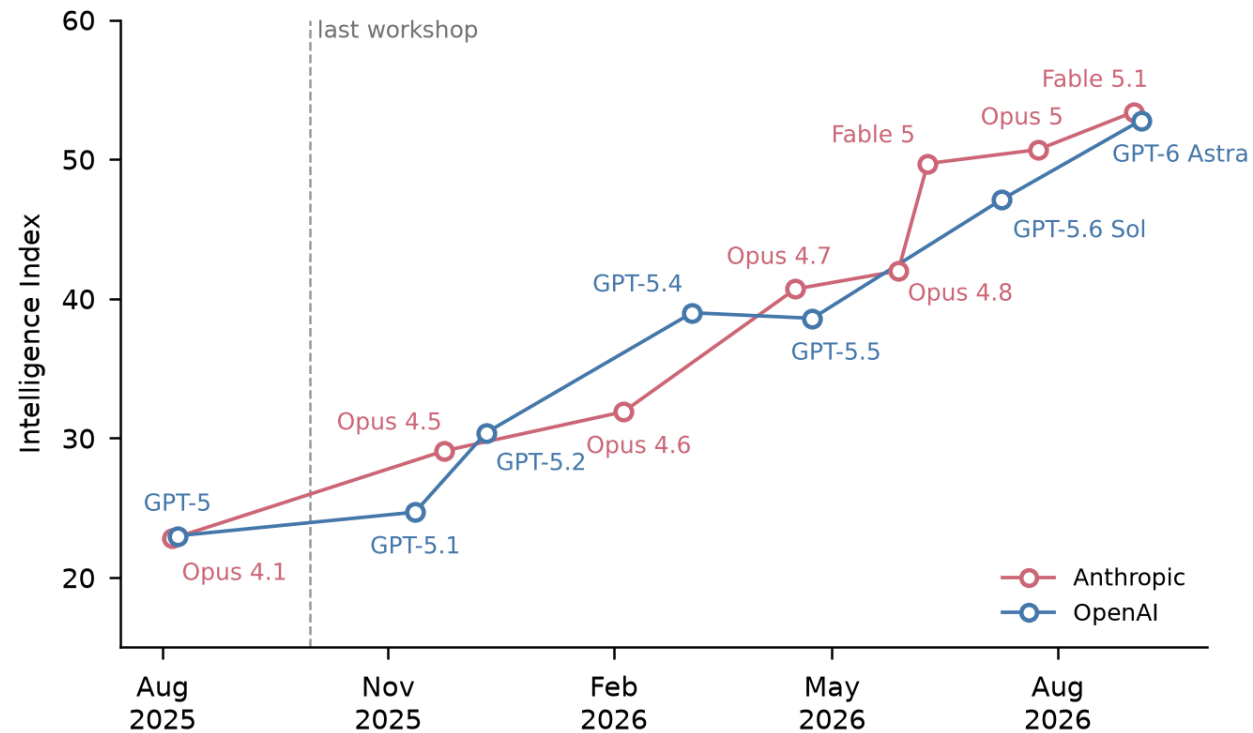
University of Freiburg

Last year the code was the hurdle



- Failing installs and unfamiliar code: a hurdle for everyone who does not code daily
- **Since then:** the models got better!

More than doubled



Artificial Analysis Intelligence Index v4.3, artificialanalysis.ai; release dates from the vendors

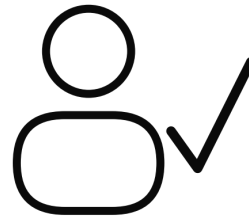
MOTIVATION

The assistant types, you decide



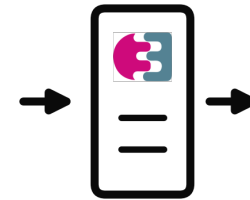
One page is enough

Instruct LLM where to find data and how to analyse it



You decide and check

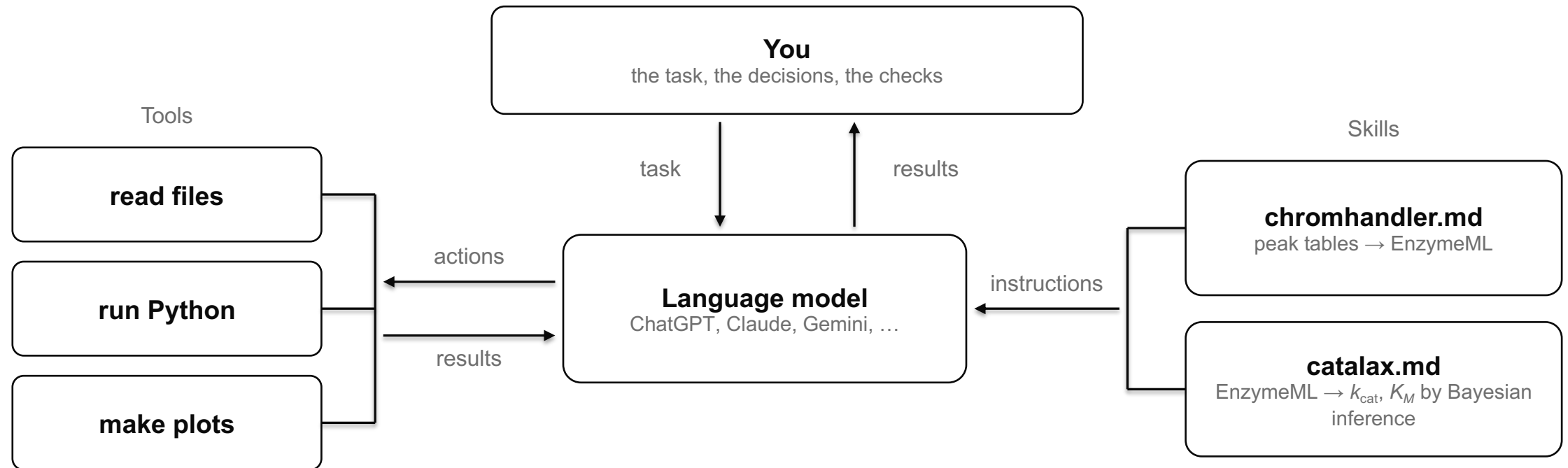
The decisions and the checks stay with the scientist; the assistant executes



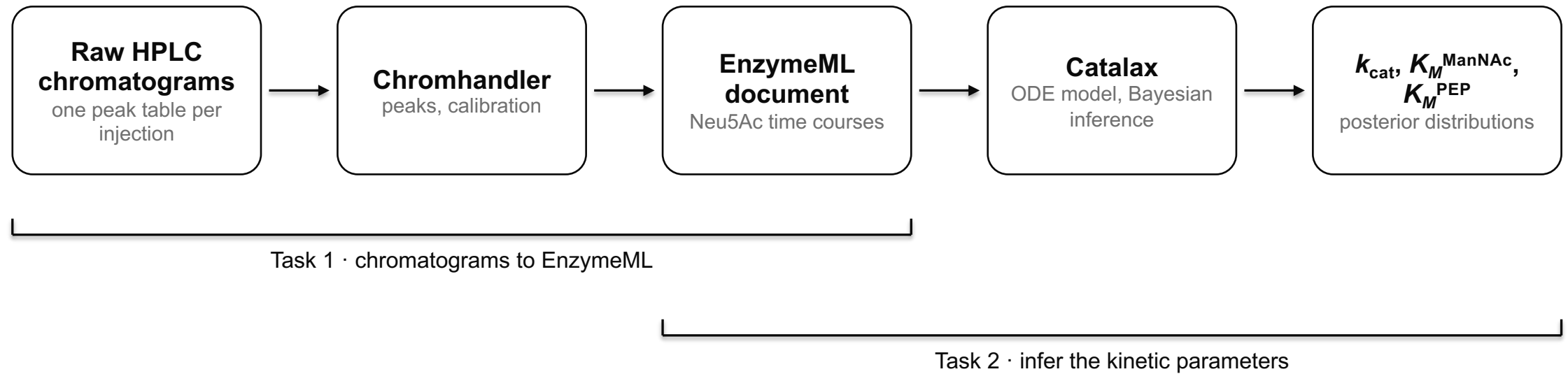
EnzymeML in between

One document carries the data between the steps. LLM writes code to do data science

Tools act, skills instruct, you decide

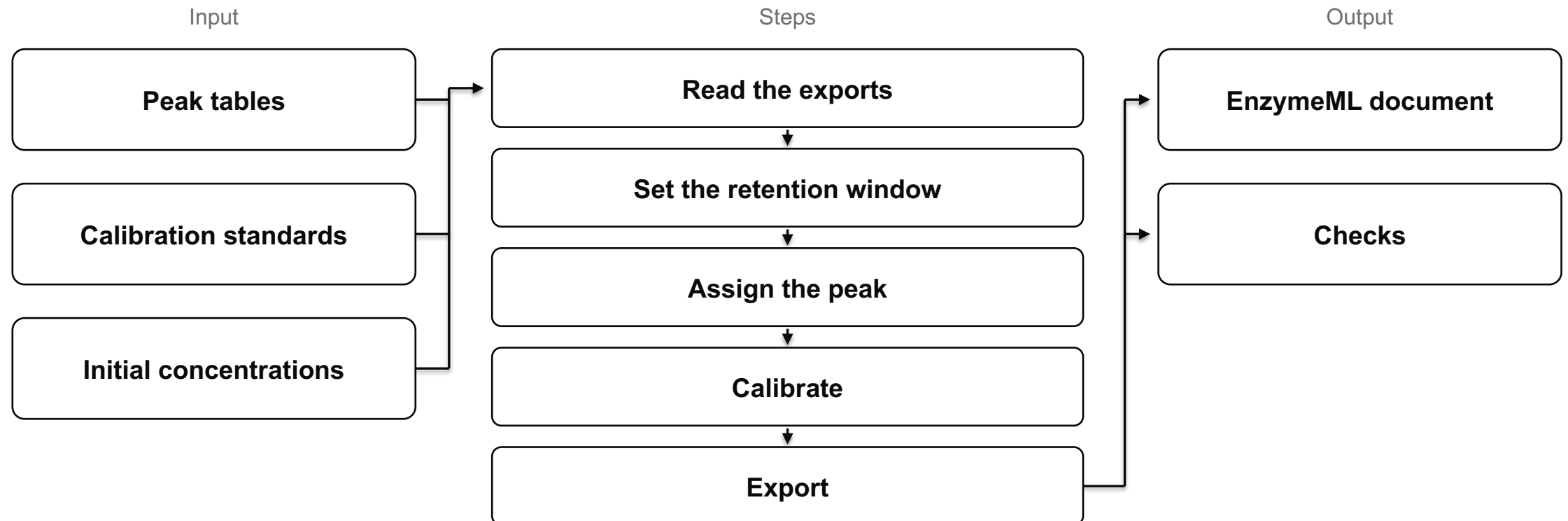


Two tasks, one document in between



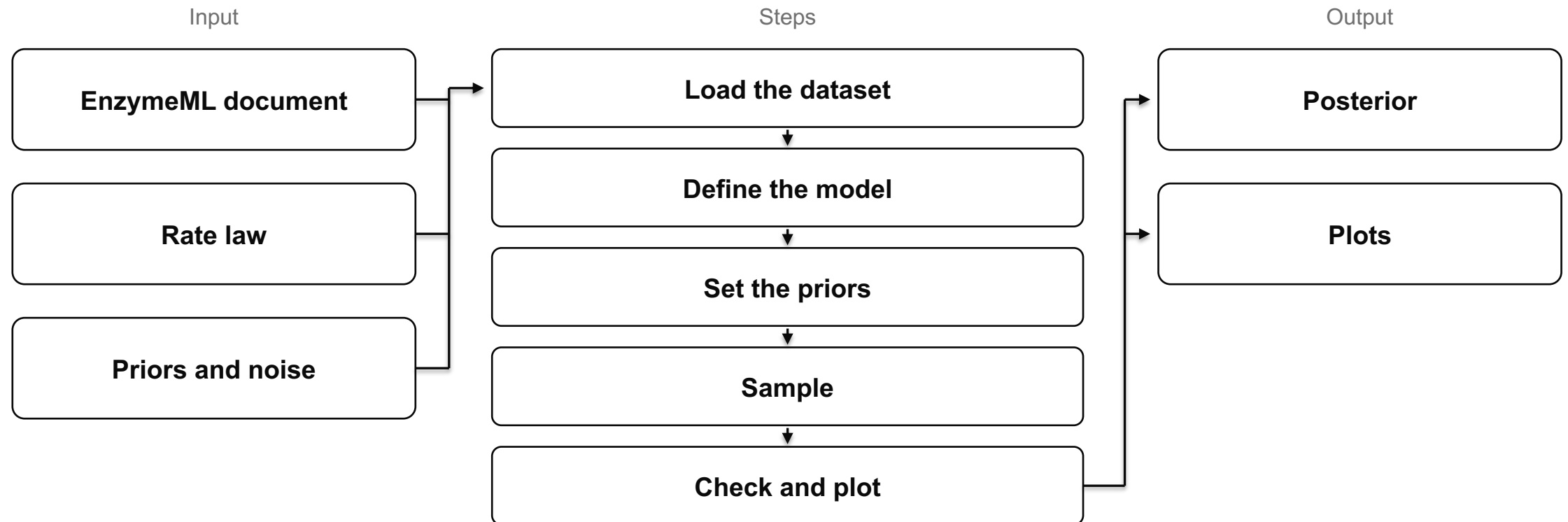
Chromhandler: from peak tables to concentrations

Reads the instrument's exports, assigns and calibrates one peak per compound, writes EnzymeML



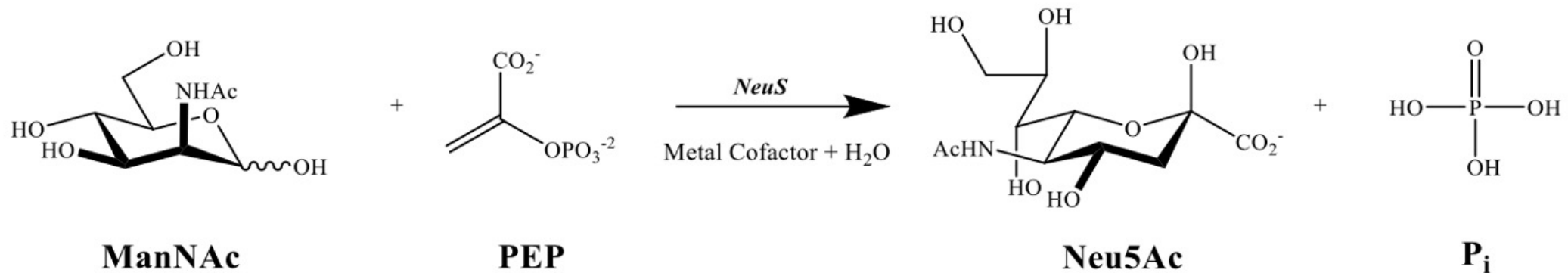
Catalax: from time courses to kinetic parameters

Builds the ODE model from a rate law and samples the posterior of its parameters



THE EXPERIMENT

ManNAc + PEP $\rightarrow$ Neu5Ac + Pi, followed by HPLC



- HPLC-PDA time-course data
- Raw data: peak tables as instrument output
- Only the product Neu5Ac is quantified

Data: Çakar et al. (2024) Kinetic characterization of two neuraminic acid synthases and evaluation of their application potential. Appl. Microbiol. Biotechnol. 108, 446 · doi:10.1007/s00253-024-13277-1

Get started: tinyurl.com/enzymeml2026

Everything is in the README

Tasks



Task 1 · chromatograms to EnzymeML: plot the data, check the assay design, calibrate, export the document



Task 2 · kinetic parameters: infer k_{cat} and K_M with their uncertainty and correlation; does the model describe the data?



At the end we discuss experiences

How to get started: choose your route



Route A · coding assistant on your computer: give it the cards in briefs/



Route B · chat assistant only: the Colab starter notebook

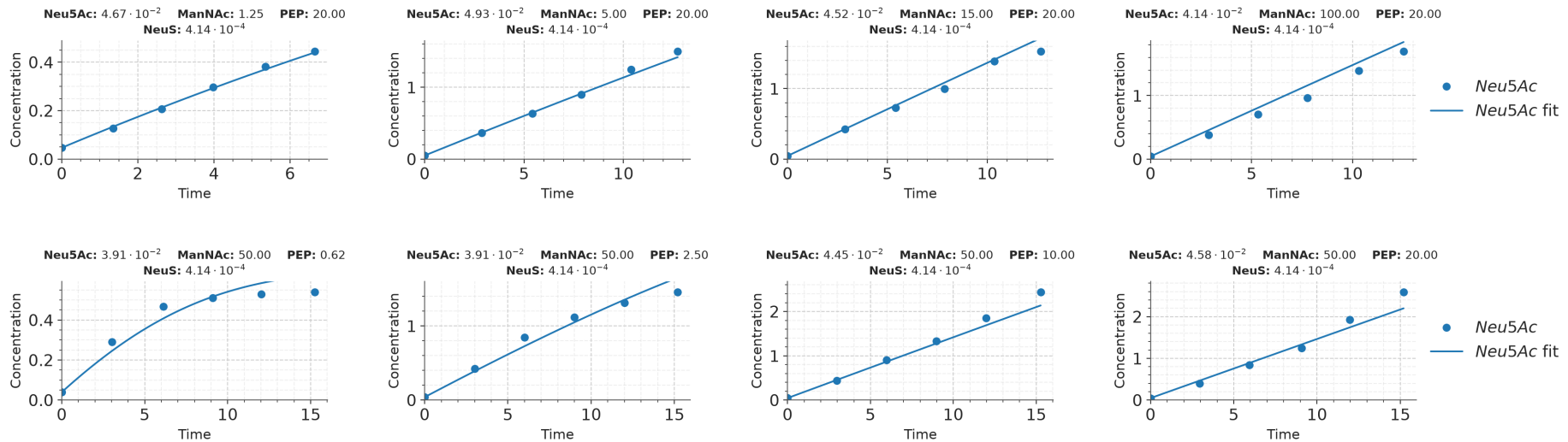


Route C · no assistant that runs code: team up



Details: README.md
sections: How to do it, Task 1, Task 2

The reference fit, and an open question



Parameter	Posterior mean $\pm$ sd
k_{cat}	$364 \pm 6 \text{ min}^{-1}$ (6.1 s^{-1} , 9.4 U/mg)
K_M^{ManNAc}	$1.53 \pm 0.19 \text{ mM}$
K_M^{PEP}	$0.59 \pm 0.07 \text{ mM}$

- One shared k_{cat} over-predicts the ManNAc series and under-predicts the PEP series
- The two series were measured on different days and differ by roughly 25 % in activity
- What would you do next?

Did it work, and what would you change?

-  **Today:** did it work? What did not? Do you already work with your data like this, for other biocatalysis problems?
-  **The cards:** did anyone read the instructions given to the language model? What would you add or remove?
-  **Documentation:** is this information in publications, in lab notebooks? How would we document experiments so that a model can act on them?
-  **Access:** which language models can you use at your university, in your group?
-  **In five years:** how are data managed and analysed? What does that imply for now?

What to take home

- A one-page card is enough for an assistant to drive our tools
- The decisions and the checks stay with the scientist
- The cards are reusable as SKILL.md or AGENTS.md in coding agents
- What did not work for you today is the most useful feedback for us

Thank you

tinyurl.com/enzymeml2026

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